SM SCHEMEMATCH

01 — INNOVATION ORIGINALITY · DIFFERENTIATION · RELEVANCE

A transparent eligibility engine, not another scheme search box.

Citizens get a ranked answer to "which schemes fit me, and why" — computed from statutory rules, with the reasoning and the missing paperwork exposed.

STEP 1
Profile capture
Form, free text, voice, or demo persona

STEP 2
Hard gates
Domicile, age, gender, income ceiling, land

STEP 3
Weighted score
Income fit, occupation, category, conditions

STEP 4
Explanation
"Why it fits" in plain language

STEP 5
Gap detection
Flags missing documents before applying

DIFFERENTIATOR

Deterministic, auditable rules

Every match is a reproducible function of 22 statutory rule sets — no hallucinated eligibility, no black-box ranking.

DIFFERENTIATOR

AI assists, it does not decide

Gemini only parses free text into a profile. The score itself is rule-based, and the app works fully with no API key.

DIFFERENTIATOR

Explains the "no" too

Ineligible schemes state which gate failed, so citizens learn the criteria instead of guessing.

Eligible people miss benefits because the rules are invisible.

Welfare information is spread across departmental portals, written in bureaucratic language, and gated by criteria that are never surfaced to the person who qualifies.

THE PROBLEM

Discovery is the bottleneck, not intent

A student, a 2-acre farmer, and a woman head of household qualify for different schemes from the same portal — and none of them can tell which without reading every rule.

WHO BENEFITS

Six citizen personas modelled end to end

College student, small farmer, homemaker and head of family, unemployed graduate, senior citizen, micro-entrepreneur — each with a one-click demo profile.

ACCESS

Kannada, Hindi, and voice

Full trilingual UI plus speech input and read-aloud output, so low-literacy and first-time internet users are not excluded by the interface.

CREDIBLE BENEFIT

Fewer rejected applications

Document-gap detection surfaces Aadhaar–NPCI seeding, Nadakacheri RD certificates and land records *before* submission — the failure points that send citizens back to the counter.

22
SCHEMES ENCODED

3
LANGUAGES

11 + 11
STATE & CENTRAL

1
SCREEN JOURNEY

03 — EXECUTION

IMPLEMENTATION DEPTH · TESTING · BUILD QUALITY

Built, tested, and deployed in one repository.

A typed React client, an Express API, a 22-scheme rule engine and a real deployment pipeline — verified end to end, not mocked.

- **Rule engine** — hard gates, weighted affinity scoring (0–100), gap detection and Markdown report generation in *matchingEngine.ts*.
- **Bilingual extractor** — regex/keyword parser handling Kannada and Devanagari numerals, used as the deterministic fallback behind Gemini.
- **Hardened AI path** — 15 s extraction and 20 s chat timeouts with graceful fallback, so a stalled model can never hang the UI.
- **18 automated tests** across matching, extraction and speech, run on the Node test runner.
- **Deployment-ready** — honours injected *PORT*, serves built assets with SPA fallback in production.

VERIFICATION

Every claim checked locally

`tsc --noEmit` clean · 18/18 tests passing · Vite production build 415 kB JS / 47 kB CSS · live Render deployment smoke-tested.

STACK

Modern, boring, maintainable

React 19 · TypeScript 7 · Vite 8 · Express 4 · Tailwind v4 · Gemini 3.8 Flash · Web Speech API · Render

6.2k	18	4
LINES OF CODE	TESTS PASSING	API ROUTES

04 — UI/UX

USABILITY · CLARITY · ACCESSIBILITY · COHERENT JOURNEY

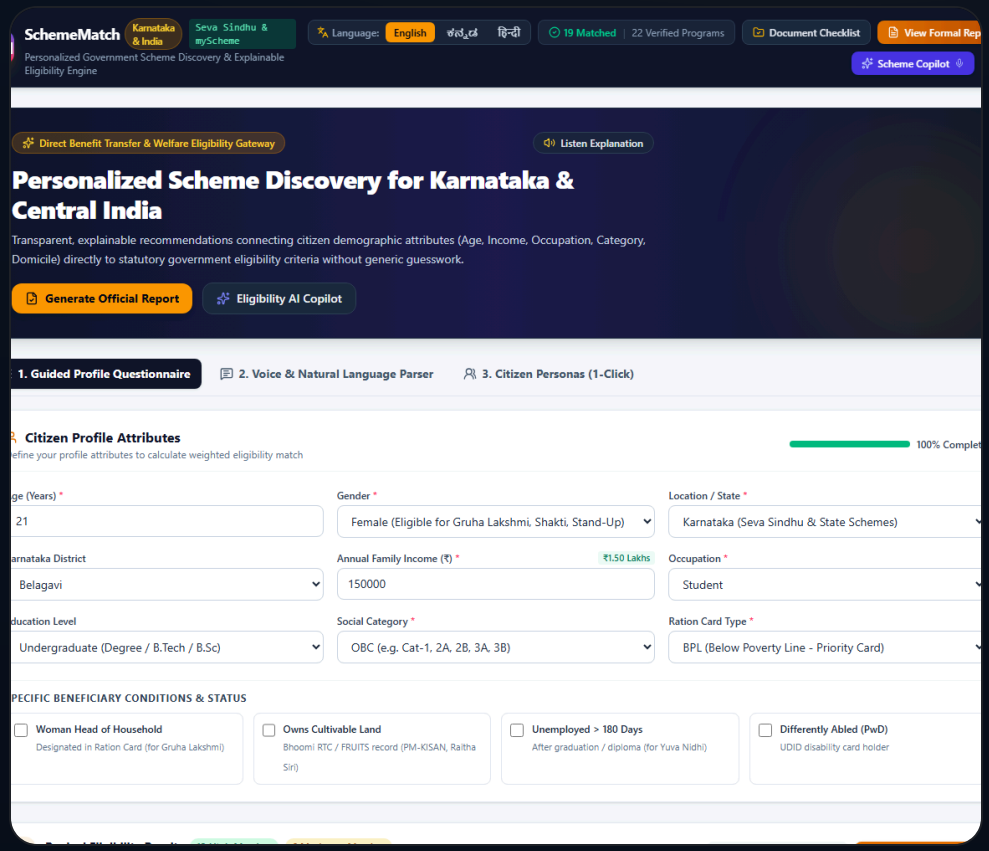
One screen from “who am I” to “apply here”.

Profile on the left, ranked matches with visible reasoning on the right — no account, no wizard, no dead ends.

- **Three input paths** — guided form, natural language, or a one-click persona, all landing in the same profile state.
- **Reasoning on the card** — match percentage, contributing factors, and the specific documents still missing.
- **Language switch that persists** — English, ಕನ್ನಡ, हिन्दी across labels, cards and generated reports.
- **Voice both ways** — speak the profile, listen to the explanation.
- **Formal report** — downloadable four-part Markdown report with the statutory disclaimer.

THE JOURNEY

Profile → Ranked matches → Why it fits → Document checklist → Official report



05 — DEMO

REAL FUNCTIONALITY · SUPPORTED BY EVIDENCE

Live in production: schememateh-codehackers.onrender.com

A 60-second path a judge can run themselves, with the expected on-screen result at each step.

STEP 1 · 10 S

Pick “Deepthi (College Student)”

A 21-year-old OBC student in Belagavi, family income ₹1.5 L. Profile, category and income populate instantly.

STEP 2 · 20 S

Read the ranked matches

SSP scholarship, Vidyasiri, AICTE Pragati and Shakti surface as high matches; each card states why and what is missing.

STEP 3 · 20 S

Open the checklist and report

Document readiness toggles live, and “View Formal Report” generates the four-part Markdown report with the disclaimer.

OBSERVED ON THE LIVE SITE

19 matched · 22 verified programs

The deployed build reports 22 schemes in its knowledge base and a complete profile score, with zero console errors and stylesheet loading confirmed.

ALSO DEMOABLE

Voice + natural language

Type “I am a 21 year old female B.Sc student from Belagavi with family income 1.5 lakh” and the extractor builds the same profile deterministically.

HONEST SCOPE

Rule engine runs client-side

On the static deployment the two AI routes are unavailable; the UI falls back to local parsing. The same repo ships a web-service blueprint that enables them.

06 — FEASIBILITY

PRACTICALITY · DEPLOYMENT · CONSTRAINTS · VIABILITY

Deployable today on a free tier, with a clear path to scale.

Static hosting for the rule engine now; a Node web service when AI endpoints are enabled. No proprietary infrastructure, no vendor lock-in.

DEPLOYMENT

Two supported paths

Static — publish *dist*, rule engine runs in the browser. **Web service** — Express + Vite build, AI routes live, health check at */api/health*.

COST

Free tier viable

Render free instance, zero database, Gemini billed only per AI call — and the app degrades to zero-cost deterministic behaviour without a key.

CONSTRAINTS, STATED PLAINLY

What it does not do yet

- No accounts or saved profiles — state is per session.
- No application submission; it links to Seva Sindhu and myScheme.
- Scheme rules are a curated snapshot and must be maintained.

TRUST

Verification is built in

Every recommendation and report carries the statutory disclaimer that approval is not guaranteed and official portals remain the source of truth.

NEXT 90 DAYS

Roadmap

- Persistence + citizen sign-in
- All 28 states and UTs
- Admin rule updates with citations
- Renewal and deadline reminders

MAINTENANCE

Data stays out of the components

Scheme criteria live in one typed dataset, so adding or correcting a scheme is a data change — not a rewrite.